A Smooth Push That Breaks a Fluid

A reader's guide to OpenAI's forced blowup for 3D Navier–Stokes, with a check of its Lean statement

Zinan Xing

Exposition only — all mathematics is due to the cited authors.

November 4, 2026

Abstract

In September 2026 OpenAI released a 166-page preprint and a Lean 4 formalization claiming that, for every viscosity $\nu > 0$, a smooth and compactly supported push applied to a fluid at rest can make the three-dimensional Navier–Stokes equations lose smoothness at time $t = 1$, while the total kinetic energy stays bounded. This settles, if correct, the two *forced* breakdown alternatives (C) and (D) of the Clay Millennium Problem; the unforced questions (A) and (B) are untouched. This guide is written for a reader who knows calculus and a little vector analysis, not fluid PDE. It explains the one idea that makes forcing powerful (*choose the flow, then read off the force*), the collapsing vortex at the heart of the construction, how tiny oscillations are made to supply exactly the push the vortex is missing, and how the remaining errors are killed to infinite order. Separately, it checks the formal Lean statements of (C) and (D) clause by clause against Fefferman's official text. Citations [OAI . . .] refer to the paper [OAI] by section or page; [background: . . .] notes give context from the wider literature.

Where things stand

- **Claimed proved** [OAI Thm. 1.1, p. 1]: for every $\nu > 0$ there exist a smooth compactly supported force and zero initial velocity for which the solution blows up at $t = 1$ with bounded energy, so no global smooth bounded-energy solution exists for that force. This is Clay alternative (C); the periodic analogue gives (D).
- **Not yet independently refereed.** The result is an author-released preprint. A Lean formalization is public; Section 9 says what it certifies.
- **Still open:** alternatives (A) and (B) — global regularity, or breakdown, *without* a force.

1 The question Clay asked

The Navier–Stokes equations describe an incompressible viscous fluid with velocity $u(x,t) \in \mathbb{R}^3$ and pressure $p(x,t)$:

$$\partial_t u + (u \cdot \nabla)u \;=\; \nu\,\Delta u - \nabla p + f, \qquad \nabla \cdot u = 0. \tag{1}$$

Read left to right: the acceleration of a fluid particle equals friction ($\nu\Delta u$, with viscosity $\nu > 0$), minus the pressure gradient, plus an external force f (gravity, a stirring paddle, ...).

In Fefferman's official problem description [2], a "physically reasonable" solution on $\mathbb{R}^3$ is one that is smooth for all $t \geq 0$ (his condition (6)) and has bounded total kinetic energy $\int |u|^2\,dx < C$ for all $t \geq 0$ (condition (7)). The data must be smooth and decay rapidly (conditions (4)–(5)). Clay asks for a proof of *any one* of four statements [2]:

	no force ($f \equiv 0$)	some smooth force allowed
whole space $\mathbb{R}^3$	(A) smooth data stay smooth forever	(C) some data and force break down
periodic box $\mathbb{R}^3/\mathbb{Z}^3$	(B) smooth data stay smooth forever	(D) some data and force break down

(A) and (B) say "the equations are well behaved"; (C) and (D) say "they are not, at least when you are allowed to push". The OpenAI paper claims (C) and (D).

Picture, not proof

Why is a force allowed at all? Because breakdown with a *smooth, rapidly decaying* force is still a genuine failure of the equations: nothing rough is fed in, yet something infinite comes out. The force is a controlled "hand" that holds the fluid on a path toward a singularity. The difficulty, as we will see, is keeping that hand smooth all the way through the moment of blowup.

2 The theorem, in words and in symbols

Theorem ([OAI Thm. 1.1, p. 1]). *For every $\nu > 0$ there are a force $f \in C_c^\infty(\mathbb{R}^3 \times (0,\infty); \mathbb{R}^3)$, a compact set $K \subset \mathbb{R}^3$ and smooth u, p on $\mathbb{R}^3 \times [0,1)$ with*

$$\partial_t u + (u \cdot \nabla)u - \nu\Delta u + \nabla p = f, \qquad \nabla \cdot u = 0, \qquad u(\cdot,0) = 0,$$

$u(\cdot,t)$ and $p(\cdot,t)$ supported in K for $0 \leq t < 1$, and

$$\sup_{0\leq t<1} \|u(t)\|_{L^2(\mathbb{R}^3)} < \infty, \qquad \limsup_{t\uparrow 1} \|u(t)\|_{L^\infty(\mathbb{R}^3)} = \infty.$$

Consequently no smooth solution on $\mathbb{R}^3 \times [0,\infty)$ with this force and zero initial velocity has uniformly bounded kinetic energy.

In words: start with still water. Stir it, smoothly, inside a bounded region and only for a while. The water stays smooth until time 1; its energy never exceeds a fixed bound; but its peak speed becomes infinite as $t \to 1$.

What the theorem does *not* say. It says nothing about water left alone ($f = 0$): that is (A)/(B), still open. It does not say turbulence causes blowup, or that weather models are wrong. It does not use rough data: the force is C^∞ with compact support in space-time, and the initial velocity is exactly zero. The construction is a single, carefully engineered flow, not a statement about typical flows.

3 The key trick: choose the flow, read off the force

Here is the idea that makes the forced problem tractable [OAI §2, p. 3; §3, p. 6]. For *any* smooth divergence-free u and any p, define the *residual*

$$R(u,p) \;:=\; \partial_t u + (u\cdot\nabla)u - \nu\Delta u + \nabla p. \tag{2}$$

Then (u,p) solves Navier–Stokes with force $f := R(u,p)$, trivially. So one may simply *design* a velocity field that blows up at $t = 1$ and declare its residual to be the force.

The catch is the word "smooth". Clay requires f to be smooth on all of $\mathbb{R}^3 \times [0,\infty)$, *including* at the blowup point $(x,t) = (0,1)$ and afterwards. A field that becomes infinite generically has a residual that becomes infinite too. So the real problem is a *cancellation problem* [OAI p. 6]: find u that blows up while every space-time derivative of $R(u,p)$ stays bounded up to $t = 1$. Everything below serves this one goal.

One viscosity is enough. If (u,p,f) works for $\nu = 1$, then $u^\nu(x,t) = \sqrt{\nu}\,u(x/\sqrt{\nu},t)$, $p^\nu = \nu\,p(x/\sqrt{\nu},t)$, $f^\nu = \sqrt{\nu}\,f(x/\sqrt{\nu},t)$ works for viscosity ν, with the same blowup time [OAI p. 6].

4 A vortex that collapses

The designed flow is built around a swirling column of fluid — a vortex — centered on the z-axis and shrinking toward the origin as $t \uparrow 1$ [OAI §2.1, pp. 3–5]. Write $\tau = 1 - t$ for the time remaining. In cylindrical coordinates (r,θ,z) with velocity components (u_r, u_θ, u_z), the leading flow is axisymmetric (independent of θ).

quantity	size in the core	source
radial width ℓ_r	$\tau^{1/2}$	[OAI p. 4]
height ℓ_z	$\tau^{1/2-h}$ (so $\ell_z/\ell_r \sim \tau^{-h} \to \infty$)	[OAI p. 4]
swirl and axial speed $\lvert u_\theta\rvert, \lvert u_z\rvert$	$\tau^{-1/2-h} \to \infty$	[OAI p. 4]
inflow speed $\lvert u_r\rvert$	$O(\tau^{-1/2})$	[OAI p. 4]
core kinetic energy	$\tau^{1/2-3h} \to 0$	[OAI p. 4]
core dissipation rate	$\tau^{-1/2-3h}$, integrable in τ iff $h < 1/6$	[OAI p. 16]

Here h is a small fixed exponent; the construction uses $0 < h < 1/100$ [OAI p. 4].

How can energy vanish while speed explodes? Energy is speed squared times volume. The core's volume is about $\ell_r^2\ell_z \sim \tau^{3/2-h}$, which shrinks faster than the speed squared τ^{-1-2h} grows; the product is $\tau^{1/2-3h} \to 0$. The blowup is a "needle" of extreme speed carrying almost no energy, which is exactly why the bounded-energy requirement (7) does not prevent it.

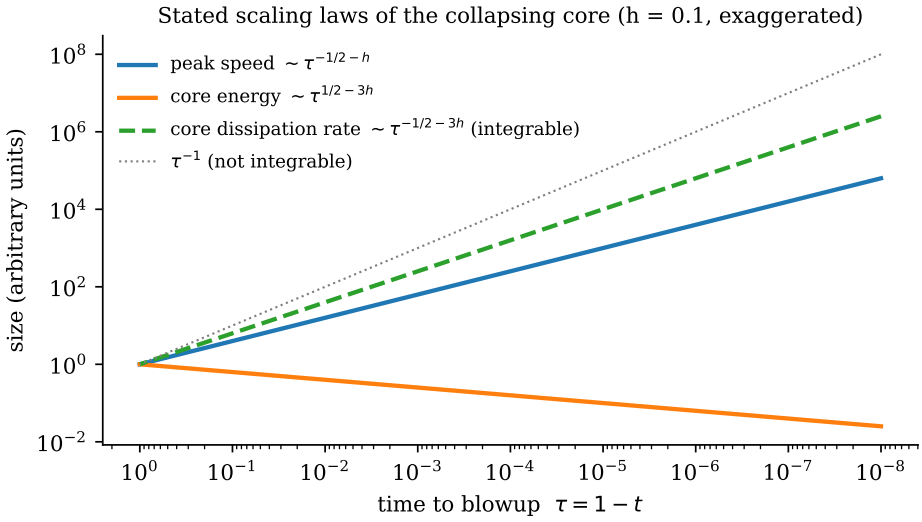


Figure 1: The stated power laws (not a simulation), with h exaggerated to 0.1 for visibility. Peak speed diverges while the core's energy tends to zero; the dissipation rate diverges, but slowly enough that its time integral is finite. The τ^{-1} line is shown for comparison.

Why spin up? For a fluid ring with no twisting force, the angular momentum $r\,u_\theta$ is conserved, so pulling fluid inward (smaller r) speeds up its swirl — the figure-skater effect [OAI p. 4]. Pressure, lowest on the axis, supplies the centripetal force; viscosity carries angular momentum outward; and fluid drawn in radially leaves through an axial outflow, which keeps the flow incompressible [OAI pp. 3–4].

Why so slender? The core is deliberately anisotropic: it shrinks like $\tau^{1/2}$ sideways but only like $\tau^{1/2-h}$ in height (Figure 2). Two consequences [OAI p. 4]. First, the swirl Reynolds number $|u_\theta|\ell_r/\nu \sim \tau^{-h}$ grows, so inertia beats friction where it matters, while the radial Reynolds number stays $O(1)$, so friction and inflow stay balanced. Second, axial diffusion is weaker than radial diffusion by a factor $\ell_r^2/\ell_z^2 \sim \tau^{2h}$; this ratio becomes the small parameter in which the whole construction is expanded [OAI §3.1].

A deliberate tilt. A vortex exactly symmetric under $z \mapsto -z$ would force $u_z(r,0,t) = 0$ and lose the radial shear of u_z near $z = 0$ that the later stages need; so the axial profile is given a small upward bias [OAI p. 5].

The precise ansatz. For readers who want the formulas: with $A = \frac{1}{2} + h$, $D = \frac{1}{2} - h$ and similarity variables $\tau = q(1-\eta^2)$, $z = q^D\eta$, $X = r^2/(2q)$, the leading fields are

$$u_\theta^{(0)} = q^{-A}E(X,\eta), \qquad u_z^{(0)} = q^{-A}U(X,\eta), \qquad r\,u_r^{(0)} = V_0(X,\eta), \qquad p^{(0)} = q^{-2A}\Pi(X,\eta)$$

[OAI (3.2) and §3.1, pp. 7–8]. The variable q is uniquely determined by (z,τ) and behaves like $\tau + |z|^{1/D}$ [OAI §3.1]: it measures "how close to the singular point" in a way that respects the anisotropy. Along the circle $z = 0$, $r = \sqrt{2X_*\tau}$ the swirl is $E(X_*,0)\,\tau^{-1/2-h} \to \infty$ [OAI §3.1].

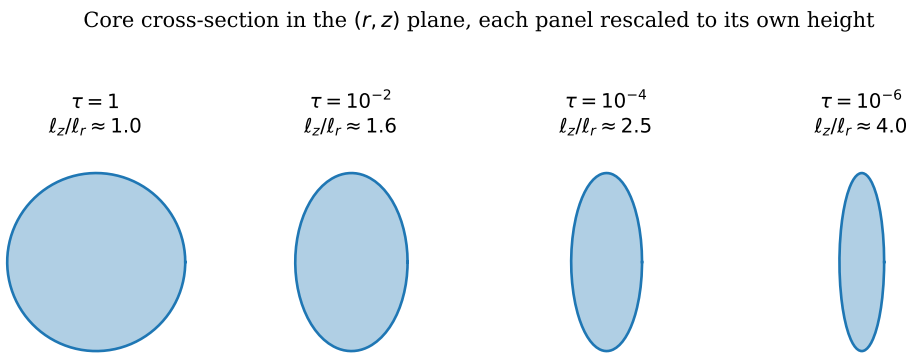


Figure 2: Core cross-sections at decreasing τ, each rescaled to its own height: the column becomes ever more slender (here $h = 0.1$, exaggerated).

5 The ring where the books don't balance

Inside the core, the profiles E, U, V_0, Π are chosen so that the leading momentum equations hold. But at the edge of the core — an annulus (ring) around the axis — the force needed to sustain this background grows without bound as $t \uparrow 1$ [OAI §2.2, p. 5]. If we stopped here, the residual (2) would blow up along with the flow, and the force would not be smooth.

The paper's first key step is to rewrite this missing momentum in a special form [OAI §3.2, pp. 9–10]. The two problematic residual components (swirl and axial) are written as the radial divergence of a *stress* $T = (T_{r\theta}, T_{rz})$ (precisely: the leading residuals with the small axial-viscosity term set aside, which is lower order in the expansion parameter [OAI (4.12) and Prop. 4.2, p. 27]):

$$R_\theta^{(0)} = -\Big(\partial_r + \frac{2}{r}\Big) T_{r\theta}, \qquad R_z^{(0)} = -\Big(\partial_r + \frac{1}{r}\Big) T_{rz},$$

and the profiles are tuned so that T vanishes except in an annulus $\sqrt{2qX_a} < r < \sqrt{2qX_b}$. For this to be possible, the residual must have zero net angular momentum and zero net axial momentum flux across radii, $\int_0^\infty r^2 R_\theta^{(0)}\, dr = 0$ and $\int_0^\infty r R_z^{(0)}\, dr = 0$ [OAI §3.2]; the paper matches five radial integrals in total in building the profiles [OAI §3.2, Prop. B.8].

Picture, not proof

A "stress" is a flux of momentum: $T_{r\theta}$ is how much swirl momentum crosses a cylinder of radius r per unit time. Writing the missing force as $-\nabla \cdot T$ turns "an unknown push is needed here" into "some momentum must be shuffled across this ring". The next section finds a way for the fluid to do the shuffling itself.

6 Making the fluid push itself

6.1 Oscillations carry momentum

Add to the background (u_B, p_B) a small rapidly oscillating field w with pressure π. The residual changes by an exact identity [OAI §3.3]:

$$R(u_B + w, p_B + \pi) = R(u_B, p_B) + \mathcal{L}_{u_B}(w, \pi) + \nabla \cdot (w \otimes w),$$

where $\mathcal{L}_{u_B}$ is the linearization at u_B. The quadratic term $\nabla \cdot (w \otimes w)$ is the key. Even though w oscillates and averages to zero, the product $w \otimes w$ does not: it has a nonzero average, the *Reynolds stress*. If that average equals the target T, the oscillations supply exactly the momentum flux that was missing.

The plane-wave toy model

Take $w = a\cos(\xi \cdot x + \varphi)$ with constant $a \perp \xi$ (the local form used in [OAI §3.3]). Then $\nabla \cdot w = 0$, the self-advection $(w \cdot \nabla)w$ vanishes identically, and the average of $w \otimes w$ over the phase is $\frac{1}{2}\, a \otimes a$. So a single wave does not disturb itself, yet carries a definite momentum flux $\frac{1}{2} a \otimes a$; slowly modulating a in space turns that flux into a force.

6.2 Why the target must lie in a cone

A wave family contributes its squared amplitude times a fixed covariance direction to the pair $(\langle w_r w_\theta \rangle, \langle w_r w_z \rangle)$. Squares are nonnegative, so with two families whose directions v_1, v_2 are independent, in this simplified picture the reachable stresses are exactly $T = c_1 v_1 + c_2 v_2$ with $c_1, c_2 \geq 0$. The paper requires $c_1, c_2 > 0$ wherever $T \neq 0$ — the *admissible stress cone* [OAI p. 10, Fig. 4]; achieving it needs an extra radial oscillation of phase $N \log X$ in the profile [OAI p. 10].

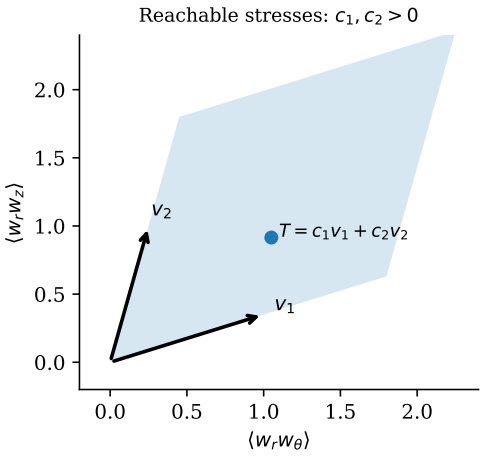


Figure 3: Two wave families with covariance directions v_1, v_2 can produce only stresses in the shaded positive cone. (Schematic; the actual v_1, v_2 vary along the annulus.)

6.3 Pulses that feed on the shear

The oscillations are organized as *pulses*: localized in radius, height and time, each wrapping around a full ring, seeded by an exponentially small force and then amplified by the background shear [OAI pp. 5–6]. For the linearized dynamics the energy of a pulse obeys [OAI §3.3]

$$\frac{d}{dt}\frac{1}{2}\int |w|^2 = -\sum_{i,j}\int w_i w_j\, \partial_j (u_B)_i \; - \int |\nabla w|^2,$$

a transfer term (energy drawn from the background shear) minus viscous dissipation. Shear also shortens the radial wavelength, which increases the damping; so each pulse first grows, then decays [OAI p. 6]. Pulse amplitudes scale as $q^{-1/2-h/2}$ and wavelengths as $q^{1/2+h/2}$ [OAI §3.3], so their squared amplitude is comparable to the background swirl divided by $q^{1/2}$, i.e. $A^2_{\text{wave}} \sim q^{-1-h}$. This is exactly the size of the target stress, $T = q^{-A-1/2}T_0 = q^{-1-h}T_0$ [OAI p. 27, Prop. 4.2].

Picture, not proof

Think of a spinning skater whose arms (the annulus) are not getting the inward pull they need. Instead of an outside hand, small ripples running around the arms are pumped up by the spin itself, and the ripples' averaged momentum flux provides the missing pull. They are placed on ever finer scales as $t \uparrow 1$.

Keeping pulses from talking to each other. Products of two *different* pulses would create uncontrolled cross terms. The paper introduces an auxiliary two-dimensional torus variable $Y \in \mathbb{T}^2$: each pulse lives on its own patch of Y, so cross products vanish identically [OAI §3.3, Lemma 6.1]; physical fields are recovered by evaluating at a phase map $Y(r,t)$.

7 Errors all the way down

After the pulses, the residual is much smaller, but not zero, and "smaller" is not enough: *every* derivative of the force must stay bounded at $(0,1)$. The paper runs an infinite correction cycle [OAI §3.4, pp. 13–15]. At stage j it adds increments $(\delta u_j, \delta p_j)$ that solve linearized amplitude equations, adjust the pulse amplitudes, invert an auxiliary-time derivative, and re-match five radial moments [OAI §3.4, Prop. 9.6]. The error after stage j is bounded by a power $q^{h\sigma_j - K_m}$ with

$$\sigma_j = \tfrac{1}{5} + \tfrac{j}{10} \to \infty$$

[OAI §3.4, p. 14], where K_m depends on the derivative order m but not on j.

Summing infinitely many corrections. Infinitely many corrections need not converge. The paper multiplies the j-th correction by a cutoff $\chi(a_j q)$ that equals one near the singular point and whose support shrinks with j [OAI §3.4]. Near $q = 0$ every term is present; away from it, only finitely many are, so the sum is locally finite for $q > 0$. [background: this is the mechanism of Borel's lemma, which builds a smooth function with any prescribed Taylor series by summing terms with shrinking cutoffs] The upshot [OAI Thm. 3.1(iii), p. 15]: for every derivative order and every N,

$$|\partial_x^\alpha \partial_t^b R(u,p)| \le C\, q^N \qquad \text{as } q \downarrow 0,$$

i.e. the residual is *flat* at the singular point — it vanishes there to infinite order.

8 Cutting it out and finishing

The far field. Beyond the annulus the flow is pure swirl, independent of height, and solves the radial heat equation for u_θ exactly, so its residual is zero there [OAI §2.3, p. 6; Thm. 3.1(iii), p. 15]. At any fixed positive radius it has smooth limits as $t \uparrow 1$.

Localization. Multiply the vector potential, the swirl and the pressure by a smooth cutoff $c = \chi_x \chi_t$ that equals one near $(0,1)$, vanishes for t near 0, and has compact spatial support; taking the curl keeps the field exactly divergence-free [OAI Prop. 10.1 and its proof, pp. 117–118]. The force is then *defined* as the residual of the cut-off fields for $t < 1$ [OAI (10.5), p. 118], so derivatives of the cutoff are simply part of the force. The growth path stays where $c = 1$, so the blowup survives [OAI (10.20)–(10.21), p. 124].

Through $t = 1$ and beyond. A force that is smooth only for $t < 1$ is not enough: Clay asks for a force on all of $[0, \infty)$. Two lemmas close this gap. First, every space-time derivative of f has a uniform limit as $t \uparrow 1$ [OAI Lemma 10.2, p. 118]: near the origin this is the flatness of Theorem 3.1(iii), and away from it the fields are smooth up to $\tau = 0$ (in the far field they are the exact heat flow). Call the limits $F_j = \lim_{t\uparrow 1} \partial_t^j f$; each F_j is smooth, supported in K, and vanishes to infinite order at the origin. Second, the force is continued past $t = 1$ by a Taylor series with shrinking cutoffs,

$$f(x, 1+\sigma) = \sum_{j \geq 0} \chi_0(b_j \sigma)\, \frac{\sigma^j}{j!}\, F_j(x),$$

with integers b_j chosen so fast-growing that every differentiated tail converges [OAI Lemma 10.3, (10.11), p. 120]. [background: this is Borel's lemma again, now used in time] The result is $f \in C_c^\infty(\mathbb{R}^3 \times (0,\infty))$ with support in $K \times [0,2]$ [OAI Lemma 10.3]. Compact support gives Clay's decay condition (5) at once [OAI p. 120].

Bounded energy comes for free. Once the force is smooth and compactly supported, the energy bound is soft. Pairing the equation with u gives $\frac{1}{2}\frac{d}{dt}\|u\|_2^2 + \|\nabla u\|_2^2 = \langle f, u\rangle$, and starting from zero velocity this yields, at viscosity one,

$$\|u(t)\|_2^2 + 2\int_0^t \|\nabla u(s)\|_2^2\, ds \;\leq\; \Big(\int_0^t \|f(s)\|_2\, ds\Big)^2 \qquad (0 \leq t < 1)$$

[OAI Lemma 10.4, (10.13)–(10.14), p. 121]. The right side is finite because f is. So the kinetic energy stays bounded and the total dissipation on $[0,1)$ is finite, with no further work. The core scalings above are consistent with this: the core energy $\tau^{1/2-3h}$ tends to zero, and the core dissipation is integrable because $h < 1/6$ [OAI p. 16].

Why no global smooth solution can exist. Suppose some smooth solution (v, P) with the same force and zero initial velocity existed on all of $[0,\infty)$, with bounded kinetic energy. The comparison lemma says that on every $[0,T]$ with $T < 1$, any smooth solution with this force, zero initial velocity and $\sup_{t\leq T}\|v(t)\|_2 < \infty$ equals u [OAI Lemma 10.5, p. 121]. Then $v = u$ on $[0,1)$; but u is unbounded as $t \uparrow 1$, while the smooth v is bounded on a compact neighborhood of $(0,1)$. Contradiction [OAI proof of Thm. 1.1, p. 124].

The comparison lemma is subtler than it looks, and this is where the precise Clay class matters: the competitor v is only assumed smooth with bounded energy. Since nothing is assumed

about v or P at infinity, the paper first recovers $\nabla\pi$, for the pressure difference $\pi = P - p$, from the velocities via Riesz transforms, then runs the energy estimate for $v - u$ on expanding balls [OAI pp. 121–123]. [background: for smooth solutions with decay, uniqueness is the classical energy argument; the work here is to avoid assuming decay]

The periodic box. Alternative (D) asks for the same breakdown on the torus $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$ [OAI Cor. 10.6, p. 125]. Because velocity, pressure and force all live in a fixed compact set, the paper shrinks the whole construction into the open unit cube and repeats it periodically [OAI pp. 125–126]:

- *Shrink.* With $\lambda > 1$ large and $t_0 = 1 - \lambda^{-2}$, set $\tilde{u}(x,t) = \lambda\, u(\lambda x, \lambda^2(t - t_0))$, $\tilde{p} = \lambda^2 p(\cdot)$, $\tilde{f} = \lambda^3 f(\cdot)$. Every term of the equation is multiplied by λ^3, so the viscosity is unchanged, and the rescaled clock $\lambda^2(t - t_0)$ reaches 1 exactly at $t = 1$.
- *Repeat.* Sum the integer translates, $U(x,t) = \sum_{k\in\mathbb{Z}^3} \tilde{u}(x + k, t)$, and likewise for pressure and force. The translates have disjoint supports with gaps between them, so at each point at most one is nonzero and even the nonlinear term $(U \cdot \nabla)U$ is just the sum of the individual ones. The pressure is periodic too, as the Clay erratum requires.
- *Uniqueness is easier here.* On the torus, integrating over the box kills both the transport and the pressure terms in the energy identity for the difference of two solutions, and Gronwall's inequality finishes [OAI p. 126].

9 What the Lean formalization certifies

9.1 How the trust is arranged

A Lean proof is only as good as the *statement* it proves. OpenAI's repository [1] uses *Comparator* [4]: a separate "challenge" file states (C) and (D) with `sorry` placeholders, and the tool checks that the project's "solution" module proves theorems whose statements, and the definitions those statements mention, are identical to the challenge's, using only the axioms `propext`, `Quot.sound` and `Classical.choice`, with a second, independent kernel (`nanoda`) enabled. The challenge statement is adapted from Google DeepMind's *Formal Conjectures* project [3].

So the trusted base shrinks to: the Lean kernel(s), the Mathlib definitions used, and *one 286-line challenge file*. Over 400,000 lines of proof in the `NavierStokes` library need not be read by a human to trust the conclusion. That makes the challenge file the place to look. A textual scan of the `NavierStokes` library finds no occurrence of `sorry`, `admit`, `native_decide` or a new `axiom` declaration. This is only a scan, not a proof check; kernel verification requires a full build, which we did not run.

9.2 Statement audit: the Lean text against Fefferman's text

Comparing the challenge file with [2] clause by clause:

No.	Fefferman's condition	Lean (structure / field)
(1)	momentum equation, $x \in \mathbb{R}^3$, $t \geq 0$	`navier_stokes`: one-sided time derivative on $[0,\infty)$; $(v \cdot \nabla)v$ as the spatial derivative applied to v; Mathlib's Laplacian (trace of the second derivative, for vector-valued maps) and gradient (via the inner product)
(2)	$\operatorname{div} u = 0$ for $t \geq 0$	`div_free`, divergence as trace of the Jacobian
(3)	$u(x,0) = u^\circ(x)$	`initial_condition`
(4)	$\lvert\partial^\alpha u^\circ\rvert \leq C(1+\lvert x\rvert)^{-K}$	`InitialVelocityConditionDecay.decay`, via the norm of the m-th derivative
(5)	$\lvert\partial_x^\alpha \partial_t^m f\rvert \leq C(1+\lvert x\rvert+t)^{-K}$	`ForceConditionDecay.decay`, joint (x,t)-derivatives within $\mathbb{R}^3 \times [0,\infty)$
(6),(11)	$p, u \in C^\infty(\mathbb{R}^3 \times [0,\infty))$	`velocity_smooth`, `pressure_smooth`
(7)	$\int \lvert u\rvert^2 dx < C$ for all $t \geq 0$	`integrable` (L^2 at each time) and `globally_bounded_energy`
(8)	u°, f periodic; p periodic (errata)	`IsOnePeriodic` fields, incl. `isOnePeriodic_pressure`
(9)	$\lvert\partial_x^\alpha \partial_t^m f\rvert \leq C(1+t)^{-K}$	`ForceConditionPeriodic.decay`
(10)	u periodic for $t \geq 0$	`isOnePeriodic_velocity`

- **Direction of strength.** (C) and (D) are *non-existence* statements, so they become *weaker* if the class of competing solutions is made smaller by extra requirements. The Lean solution class asks for (1), (2), (3), (6), (7) [resp. (1), (2), (3), (10), (11) and periodic pressure], and nothing more. The extra field `integrable` ($u(\cdot,t) \in L^2$) does not shrink the class: for smooth fields it follows from (7). Its role is technical — in Lean, the integral of a non-integrable function is defined to be 0, so without it the energy bound could hold vacuously.
- **Junk values.** Divergence and spatial derivatives are defined through `fderiv`, which returns 0 at non-differentiable points. This is harmless here because the solution class requires C^∞ smoothness.
- **Quantifiers.** Both theorems are stated for every $\nu > 0$, matching "for $\nu > 0$" in [2].

We found no discrepancy between the formal and the official statements. This is a reading-level check by the author, not reviewed by a Lean expert.

9.3 What Lean does not certify

A successful Comparator run certifies that the formal statements (C) and (D) follow from Mathlib and the listed axioms. It does not certify that the 166 pages of prose are correct as written, or that the informal Theorem 1.1 is literally what was formalized (the formal objects are the Comparator statements above). Whether the result counts for the Clay Prize is for the Clay Institute to decide.

10 What remains open

- **(A) and (B)**: does water left alone, starting smooth, stay smooth forever? Untouched by this work.

[background: prior milestones, surveyed in [OAI §1.1, pp. 2–3], include Leray's weak solutions, partial regularity of Caffarelli–Kohn–Nirenberg, Tao's blowup for an averaged Navier–Stokes equation, and nonuniqueness of weak solutions by convex integration]

(Prepared with the assistance of Claude, under the direction of the author. Released under CC BY 4.0.)

References

[OAI] OpenAI, *Finite time blowup for Navier–Stokes*, preprint, September 2026, https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf.

[1] OpenAI, *NavierStokesAndEuler* (Lean 4), commit `f9e8bc5b`, https://github.com/openai/NavierStokesAndEuler.

[2] C. L. Fefferman, *Existence and smoothness of the Navier–Stokes equation*, Clay Mathematics Institute official problem description, with errata, https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf.

[3] Google DeepMind, *Formal Conjectures*, `FormalConjectures/Millennium/NavierStokes.lean`, https://github.com/google-deepmind/formal-conjectures.

[4] Lean FRO, *Comparator*, https://github.com/leanprover/comparator.